Unit 5-Congestion Control

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What Is Congestion?

Congestion occurs when the number of packets being transmitted through the network approaches the packet handling capacity of the network

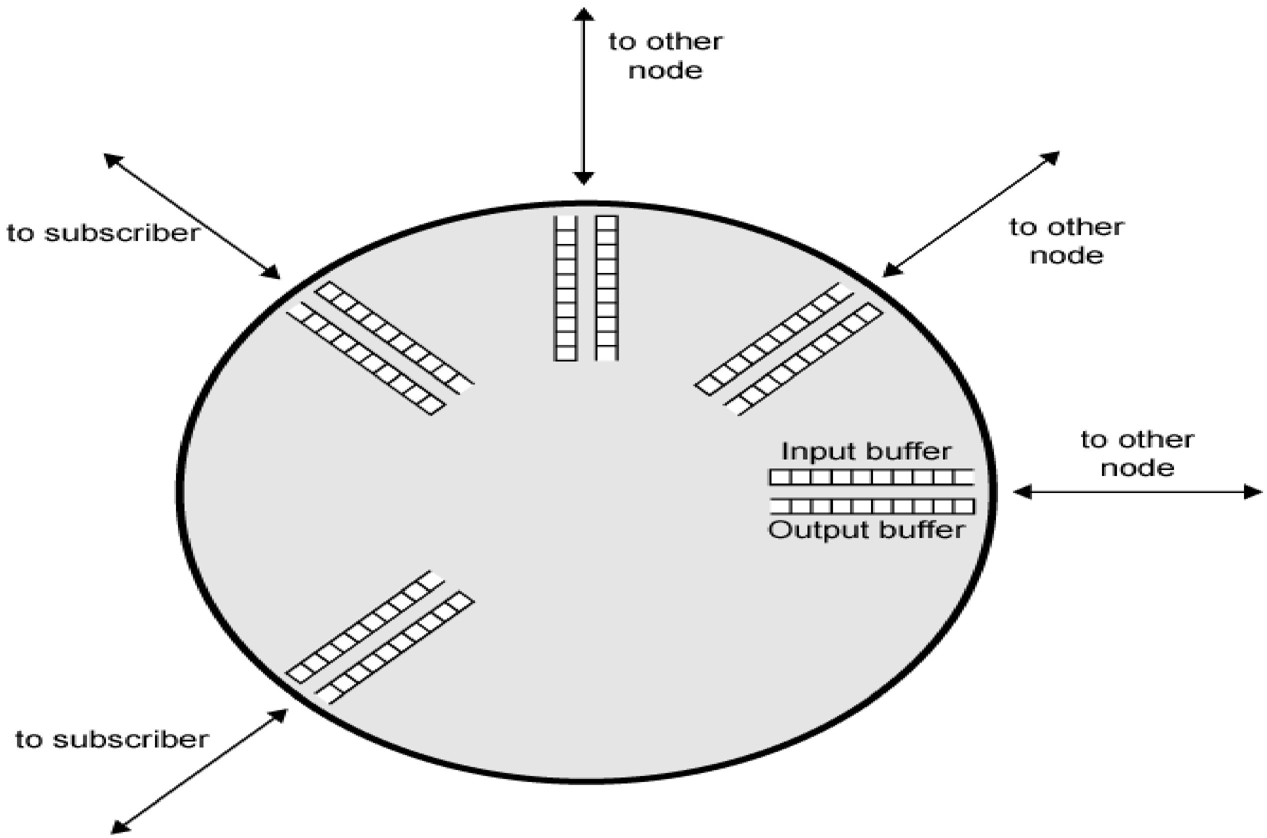
Congestion control aims to keep number of packets below level at which performance falls off dramatically

Data network is a network of queues

Generally 80% utilization is critical

Finite queues mean data may be lost

Queues at a Node



Effects of Congestion

Packets arriving are stored at input buffers

Routing decision made

Packet moves to output buffer

Packets queued for output transmitted as fast as possible

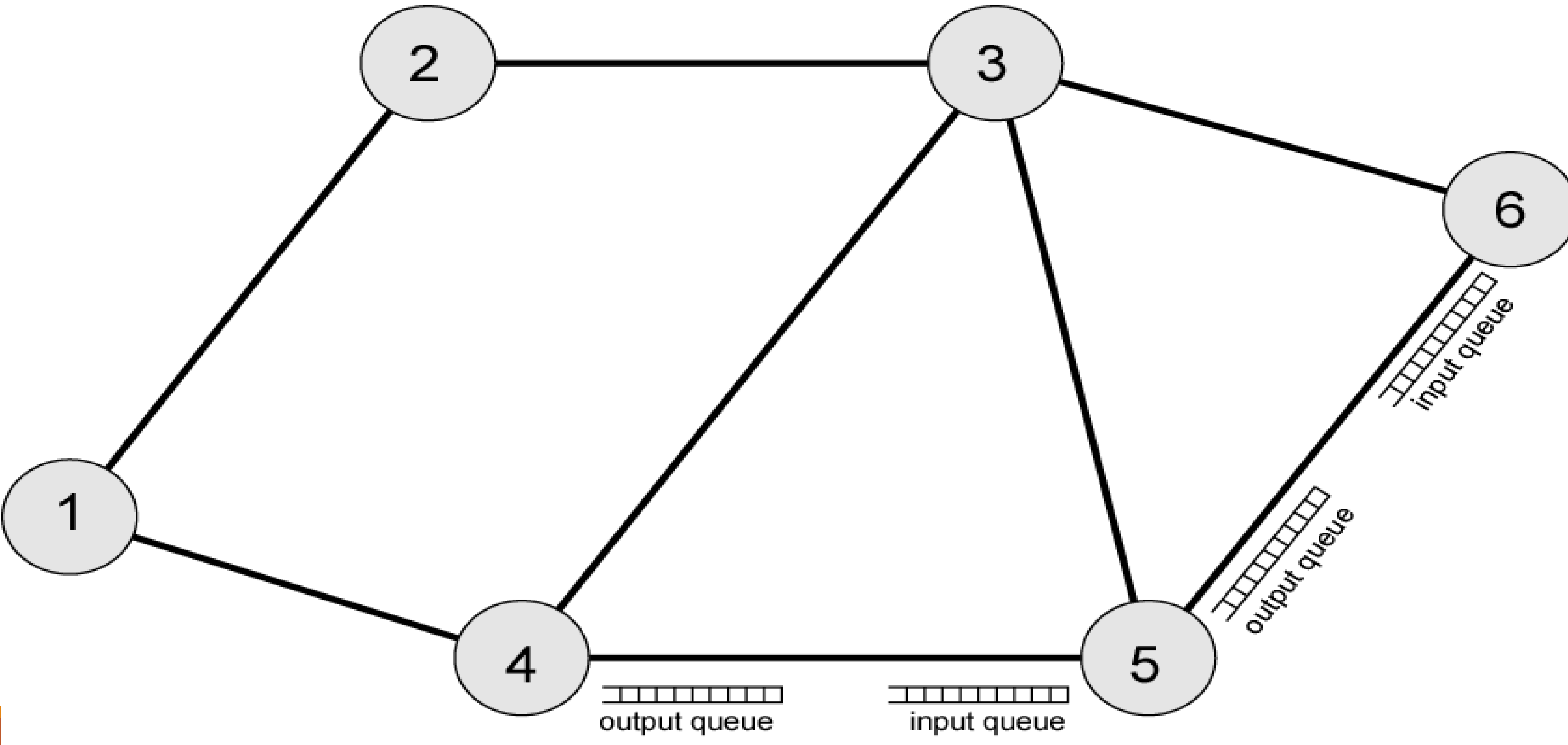
- Statistical time division multiplexing

If packets arrive to fast to be routed, or to be output, buffers will fill can discard packets

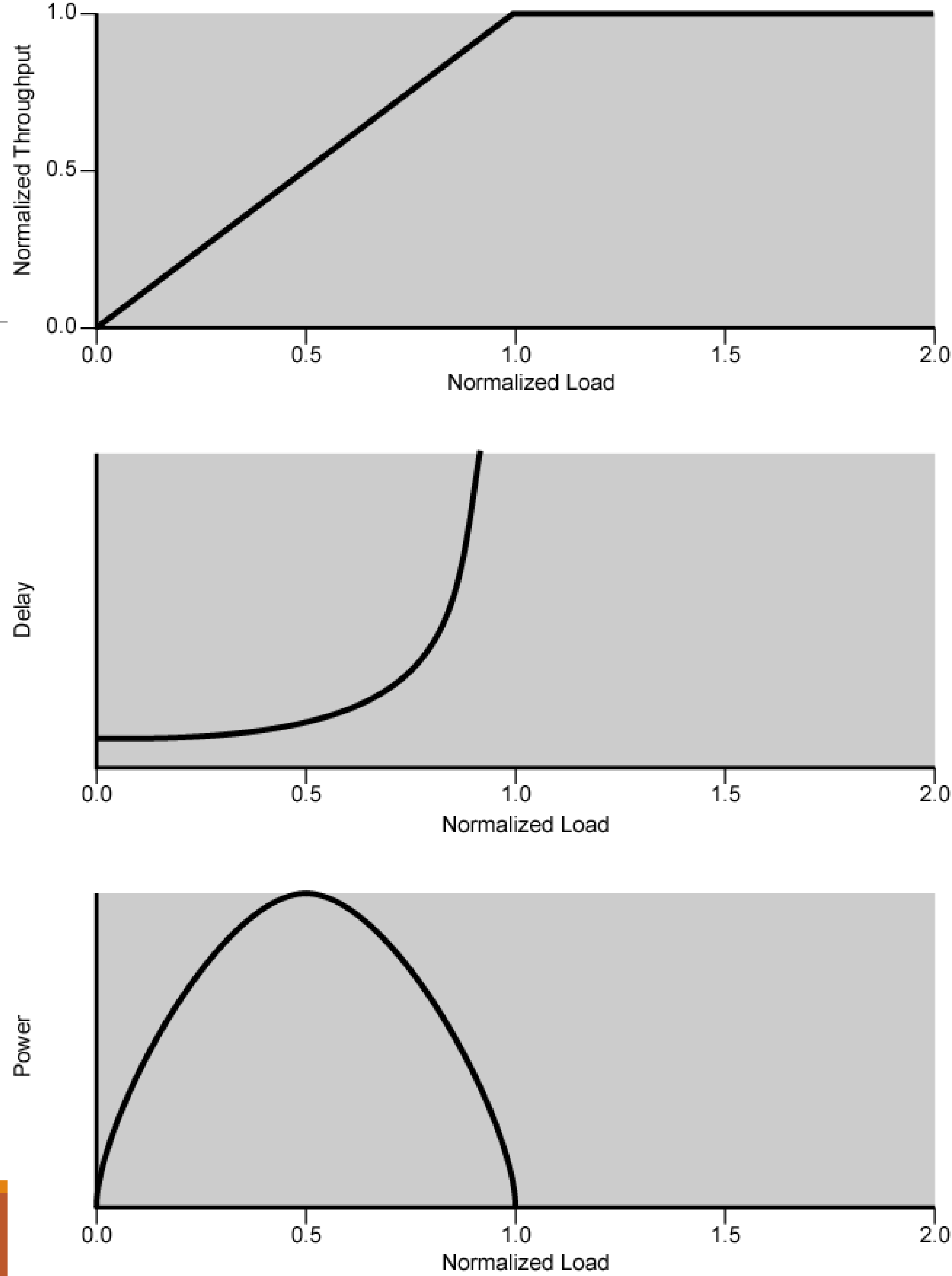
Can use flow control

- Can propagate congestion through network

Interaction of Queues



Ideal Network Utilization



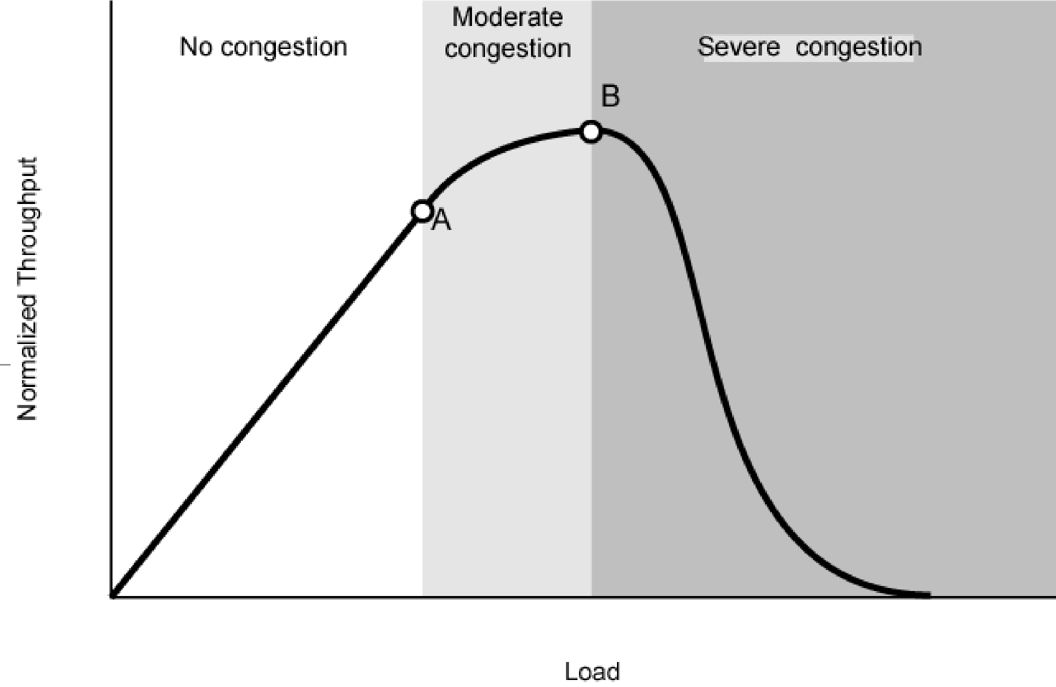
Practical Performance

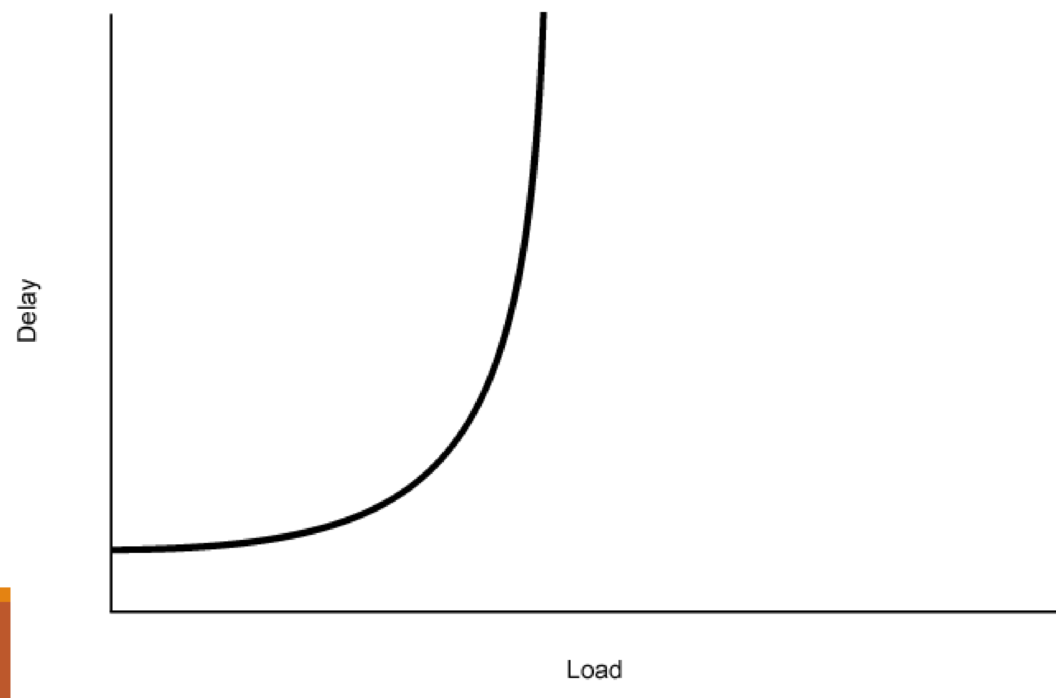
Ideal assumes infinite buffers and no overhead

Buffers are finite

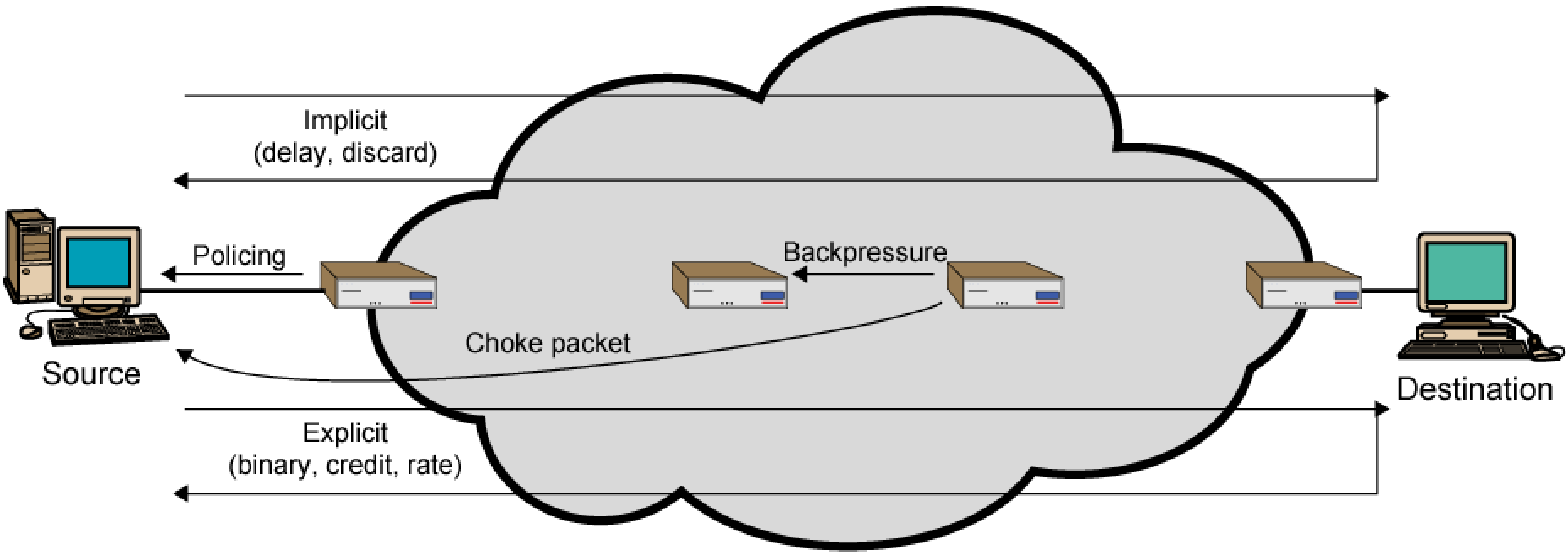
Overheads occur in exchanging congestion control messages

Effects of Congestion - No Control





Mechanisms for Congestion Control



Backpressure

If node becomes congested it can slow down or halt flow of packets from other nodes

May mean that other nodes have to apply control on incoming packet rates

Propagates back to source

Can restrict to logical connections generating most traffic

Used in connection oriented that allow hop by hop congestion control (e.g. X.25)

Not used in ATM nor frame relay

Only recently developed for IP

Choke Packet

Control packet

- Generated at congested node
- Sent to source node
- e.g. ICMP source quench
  - From router or destination
  - Source cuts back until no more source quench message
  - Sent for every discarded packet, or anticipated

Rather crude mechanism

Implicit Congestion Signaling

The source itself acts pro-actively and tries to d e c r e a s e the existence of congestion by making local observations

Transmission delay may increase with congestion

Packet may be discarded

Source can detect these as implicit indications of congestion

Useful on connectionless (datagram) networks

- e.g. IP based
  - (TCP includes congestion and flow control - see chapter 17)

Used in frame relay LAPF

Explicit Congestion Signaling

Network alerts end systems of increasing congestion

special packets are sent back to the sources to curtail down the congestion

End systems take steps to reduce offered load

Backwards

- Congestion avoidance in opposite direction to packet required

Forwards

- Congestion avoidance in same direction as packet required

Categories of Explicit Signaling

Binary

- A bit set in a packet indicates congestion

Credit based

- Indicates how many packets source may send
- Common for end to end flow control

Rate based

- Supply explicit data rate limit
- e.g. ATM

Traffic Management

Fairness

Quality of service

- May want different treatment for different connections

Reservations

- e.g. ATM
- Traffic contract between user and network

Congestion Control in Virtual-Circuit Subnets

- **Congestion is dynamically controlled in virtual-circuit subnets and includes the following techniques**

1. **Admission control**
2. **Select alternative routes**
3. **Negotiate level of service**

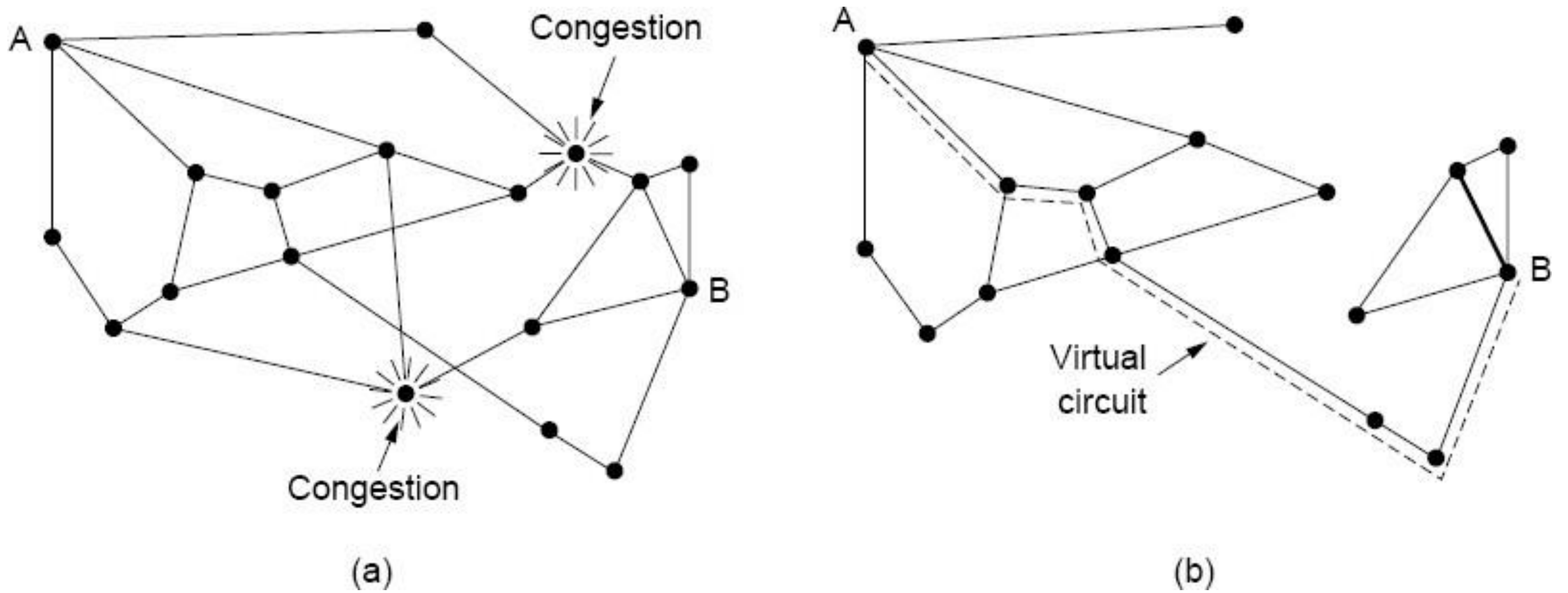
Congestion Control in Virtual-Circuit Subnets

1. Admission control. :

- Once congestion has been signaled, no more virtual circuits are set up until the problem has gone away.
- Thus, attempts to set up new transport layer connections fail.
- telephone system also uses admission control.

Congestion Control in Virtual-Circuit Subnets

2. **selecting alternative route** : allow new virtual circuits but carefully route all new virtual circuits around problem area.



(a) A congested subnet. (b) A redrawn subnet that eliminates the congestion. A virtual circuit from A to B is also shown.

Congestion Control in Virtual-Circuit Subnets

3. Negotiate level of service

- This host normally specifies the volume and shape of the traffic, quality of service required, and other parameters.
- the subnet will agree to this and typically reserve required resources along the path when the circuit is set up
- These resources can include table and buffer space in the routers and bandwidth on the lines.

Congestion Control in Packet Switched Networks

Send control packet to some or all source nodes

- Requires additional traffic during congestion

Rely on routing information

- May react too quickly

End to end probe packets

- Adds to overhead

Add congestion info to packets as they cross nodes

- Either backwards or forwards

Congestion Control in

Datagram Subnets

Congestion Control in Datagram Subnets

- For each output line, router maintains an estimate U of its
- utilization: $u_{new} = au_{old} + (1 - a) f$

  a □ a constant.

  f (0 or1) □ a sample of the instantaneous line utilization, made periodically.

If u_{new} > threshold value , the output line enters a warning state.

- If a packet's output line is in warning state, some action is taken

  1. The warning bit
  2. Choke packets

1. The Warning Bit

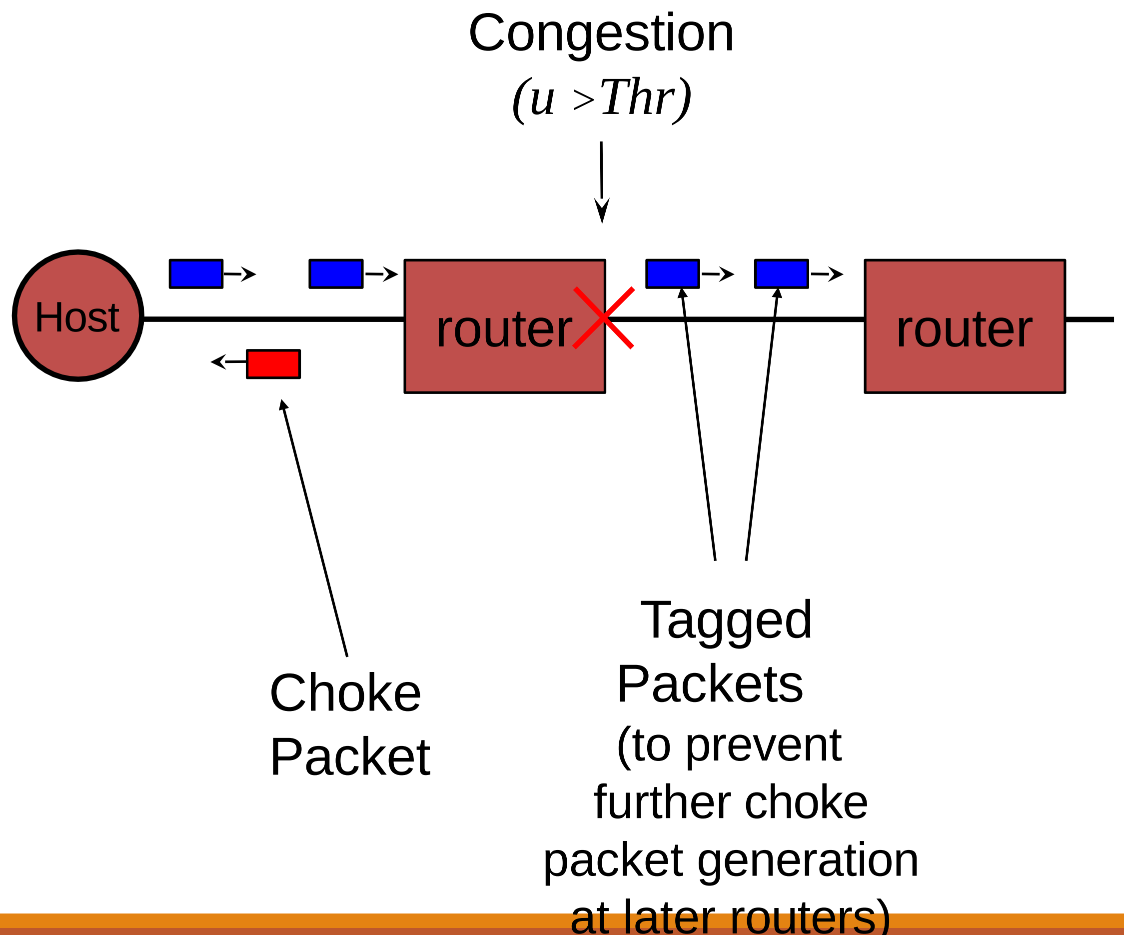
1. Router sets a *special bit* in the packet's header.
2. Destination copies the bit into the next ACK and sends to the source.
3. The sender monitors the number of ACK packets it receives with the warning bit set and adjusts its transmission rate accordingly.
4. As long as the warning bits continue to flow in, the source continues to decrease its transmission rate.

2. Choke Packets

- The router in trouble sends a choke packet back to the source host, giving it the destination found in the packet.
- The original packet is tagged so that it will not generate any more choke packets.
- When the source gets a chock packet, it cuts rate by half, and ignores further choke packets coming from the same destination for a fixed period
- After that period has expired, the host listens for more choke packets. If one arrives, the host cut rate by half again. If no choke packet arrives, the host may increase rate

2. Choke Packets

The original packet is tagged so that it will not generate any more choke packets.



2. Choke Packets - example

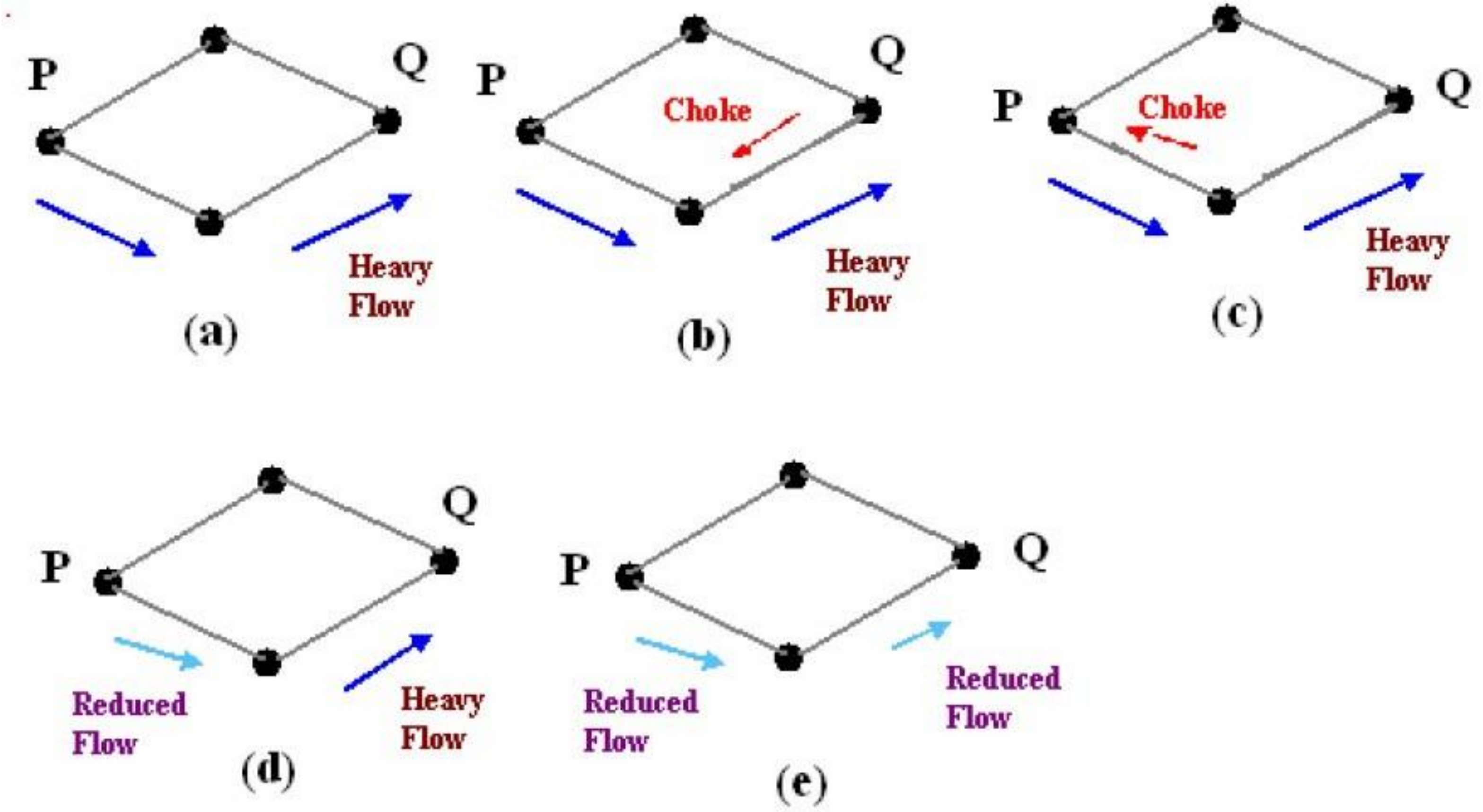
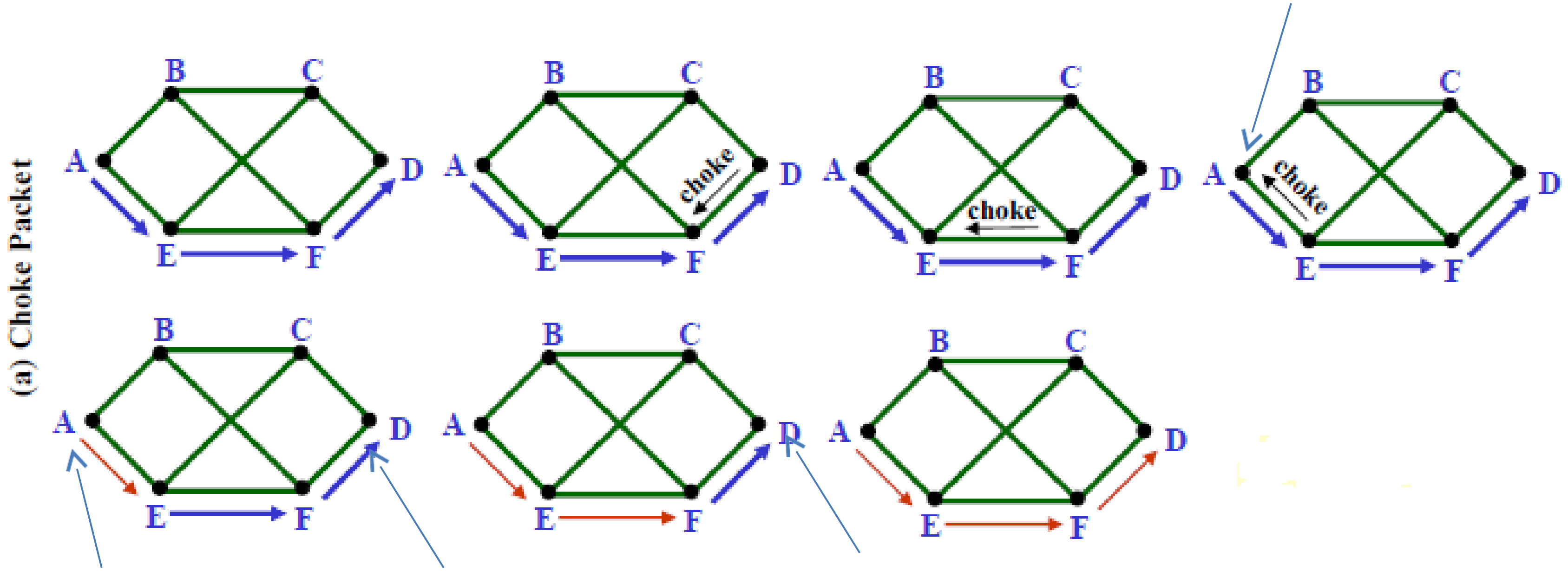


Figure Depicts the functioning of choke packets, (a) Heavy traffic between nodes P and Q, (b) Node Q sends the Choke packet to P, (c) Choke packet reaches P, (d) P reduces the flow and send a reduced flow out, (e) Reduced flow reaches node Q

Problem with Choke Packet method

- At high speeds over long distances, this method does not work well because by the time choke packet reach the source, already a lot of packets destined to the same original destination would be out from the source.

Choke packet has reached A

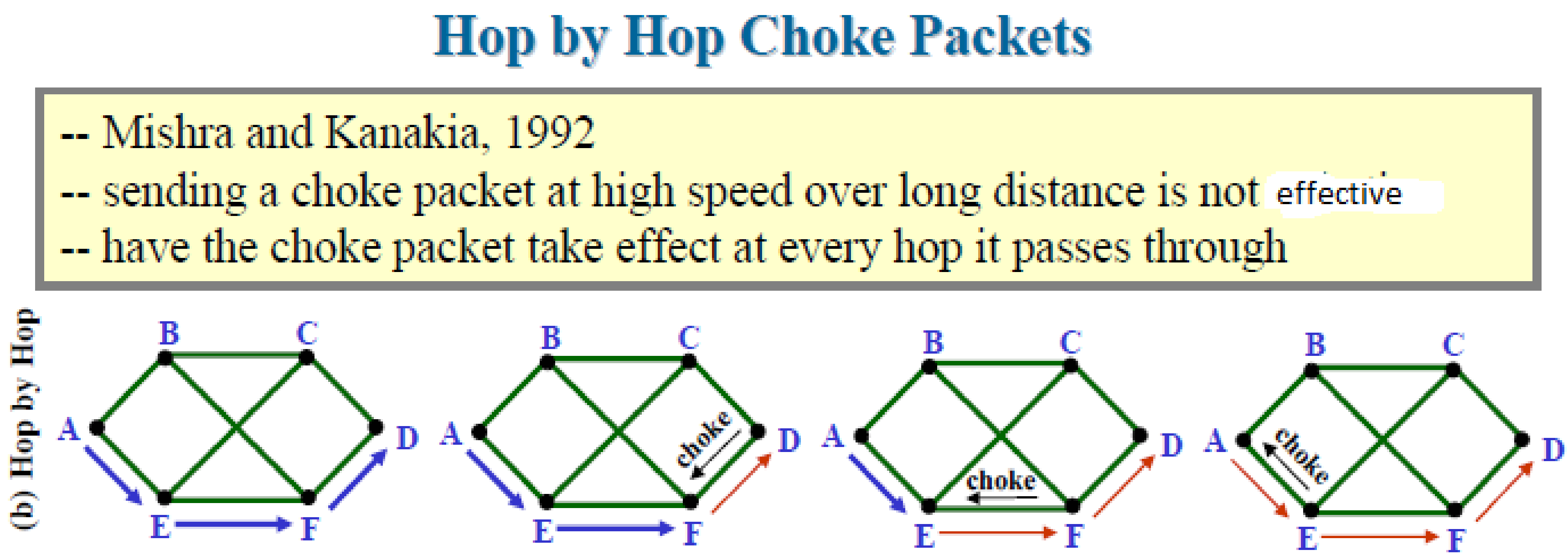


A reduces its traffic

But D is receiving data at high speed

D is still receiving data at high speed

Hop-by-Hop Choke Packets



As soon as the choke packet reaches F, F is required to reduce the flow to D. Doing so will require F to devote more buffers to the flow, since the source is still sending away at full blast, but it gives D immediate relief

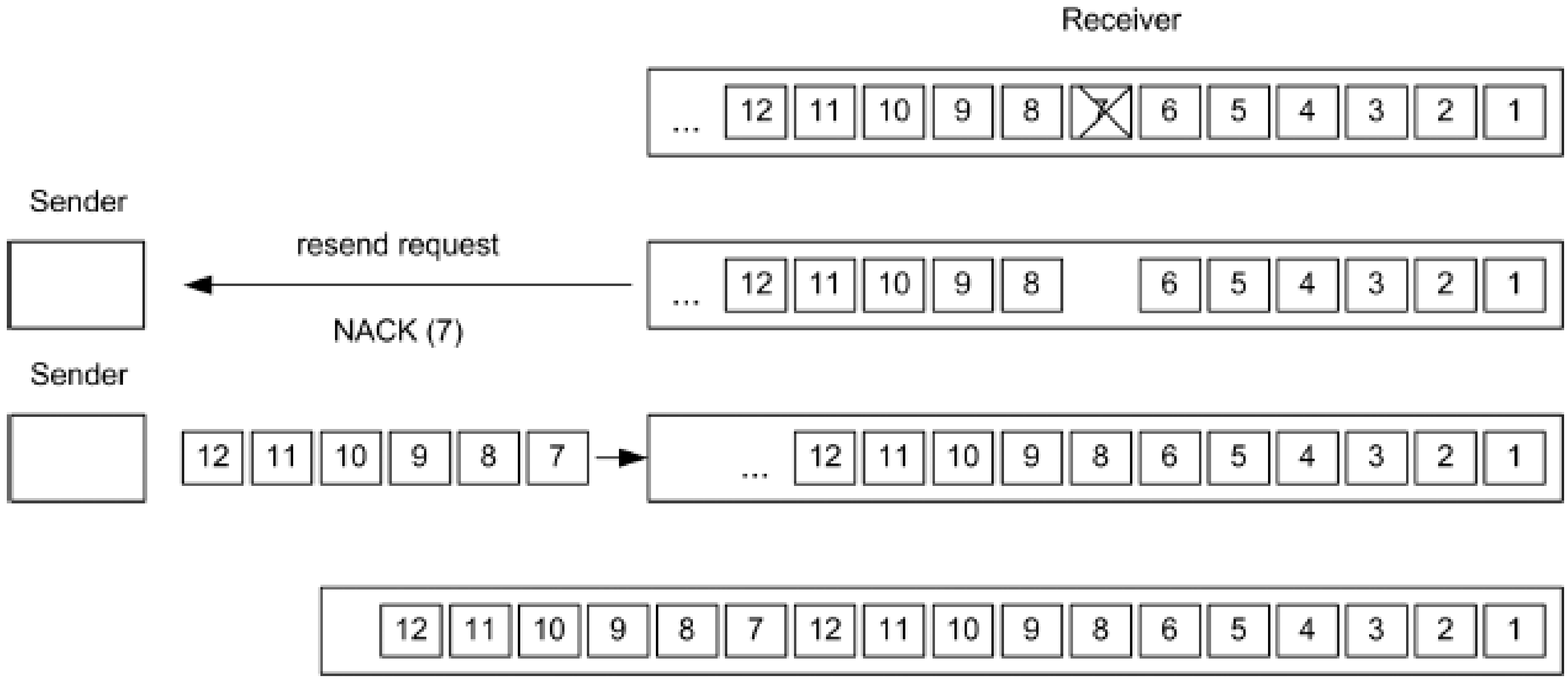
3. Load Shedding

A router , when heavily loaded and none of the methods work , can just pick packets drop. This is known as **Load Shedding .**

- **Approaches to discard packets**
  - Select packets at random
  - Based on application
    - For File Transfer : keep old, discard new
    - For multimedia : keep new , discard old
  - Intelligence discard policy

3. Load Shedding

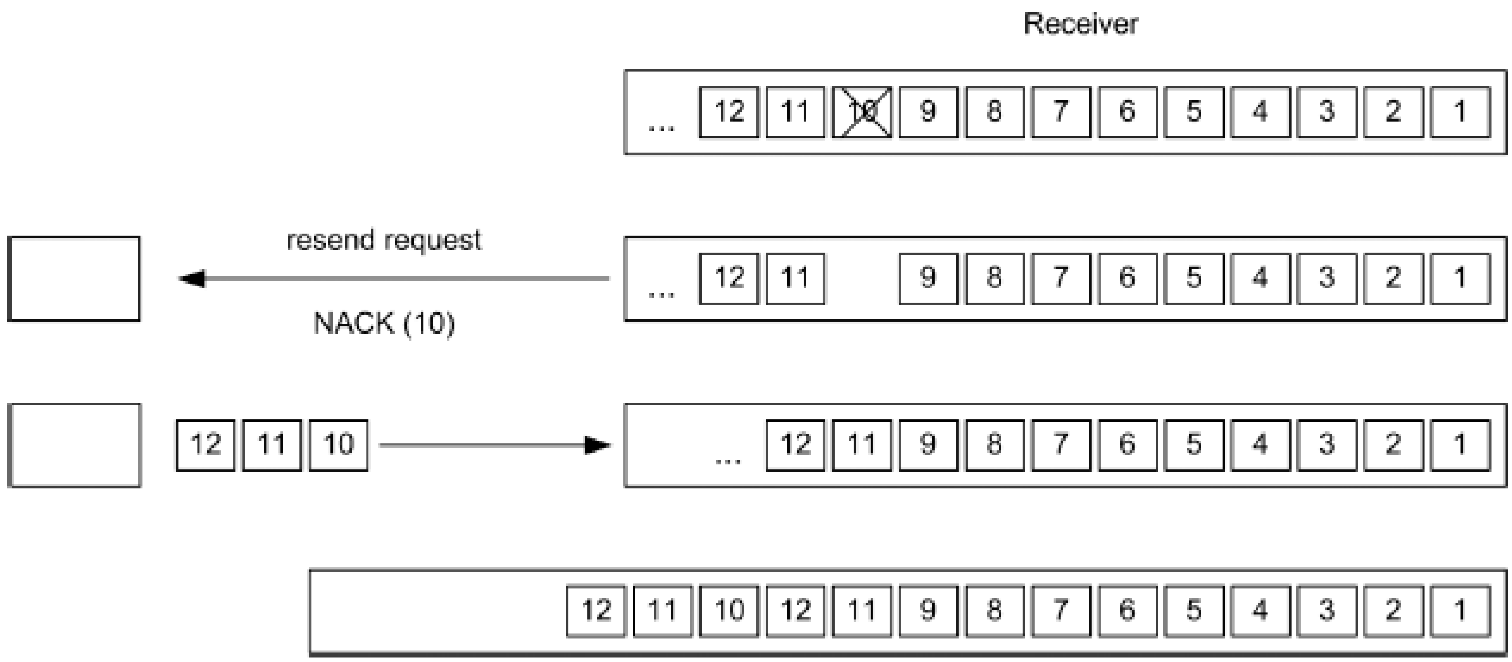
Discarding old packets



Discarding packet 7 provokes retransmission of these six packets

3. Load Shedding

Discarding new packets



Alternatively, the router can discard packet 10 so that only three packets have to be resend

3. Load Shedding

To implement **intelligent discard policy**,

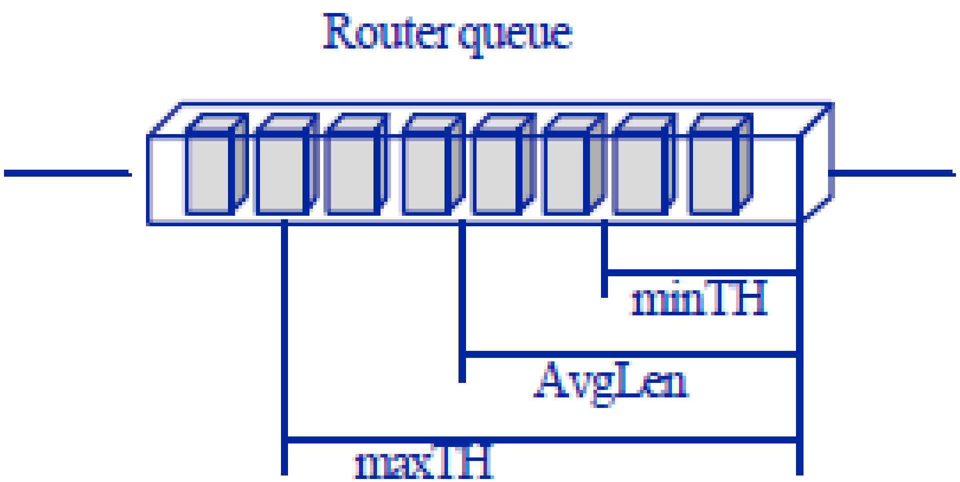
1. For this , applications must mark their packets in priority classes
2. When packets have to be discarded, routers can first drop packets from the lowest class packets, then the next lowest class, and so on

4. Random Early Detection (RED)

- **A load shedding technique .**
- idea is to avoid congestion rather than react to it.
- i.e. Discard packets before all buffer space is exhausted
- drop packets before the situation has become hopeless.

Random Early Detection (RED)

- Routers maintain a running average of their queue lengths . When the average queue length on some line exceeds a threshold, the line is said to be congested and action is taken.



Random Early Detection (RED)

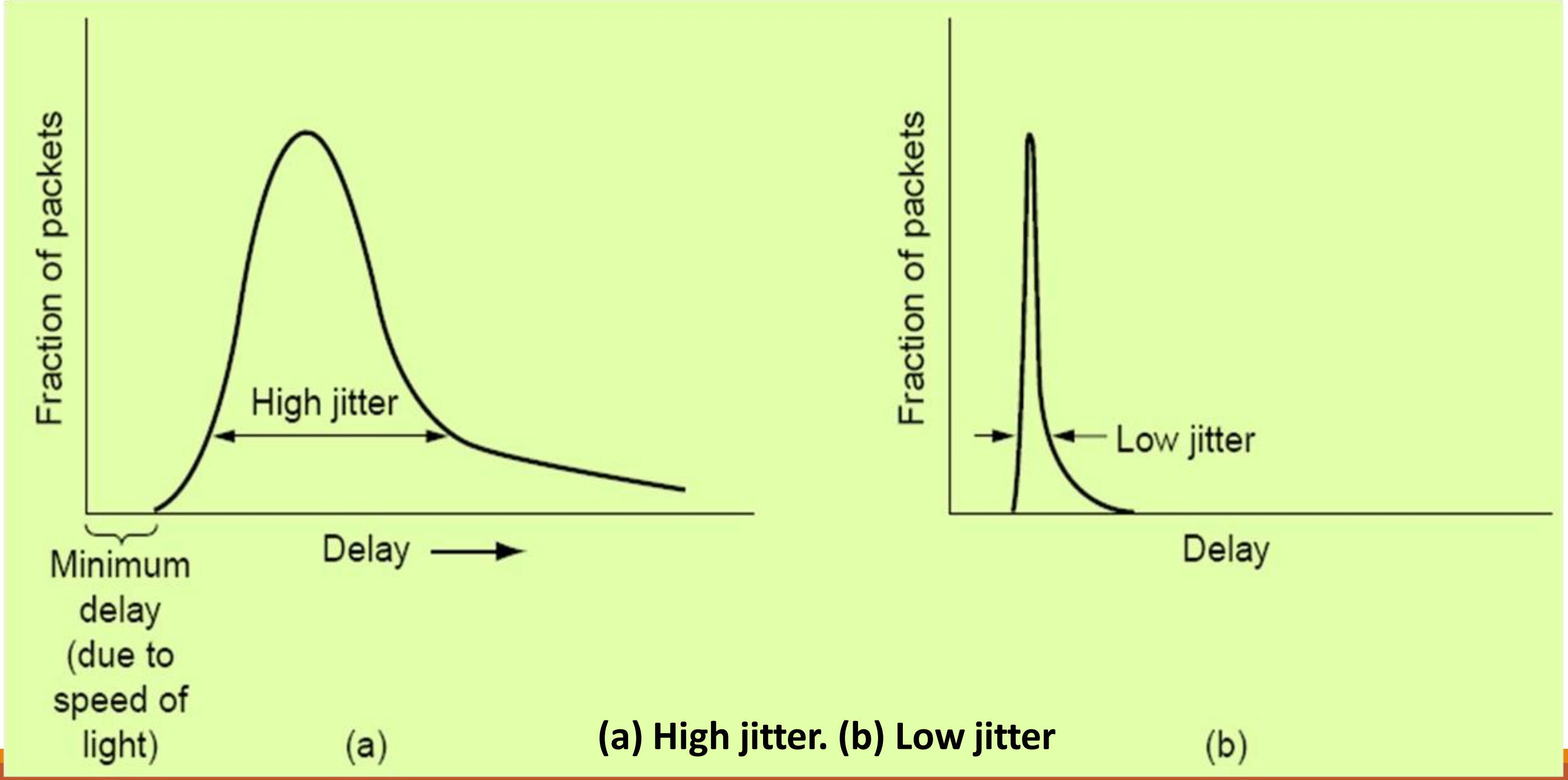
How should the router tell the source about the problem?

1. send it a choke packet
   - puts even more load on the already congested network

2. Discard the selected packet and not report it
   - When a source notices the lack of ACK, it reduces transmission rate.

5. Jitter Control

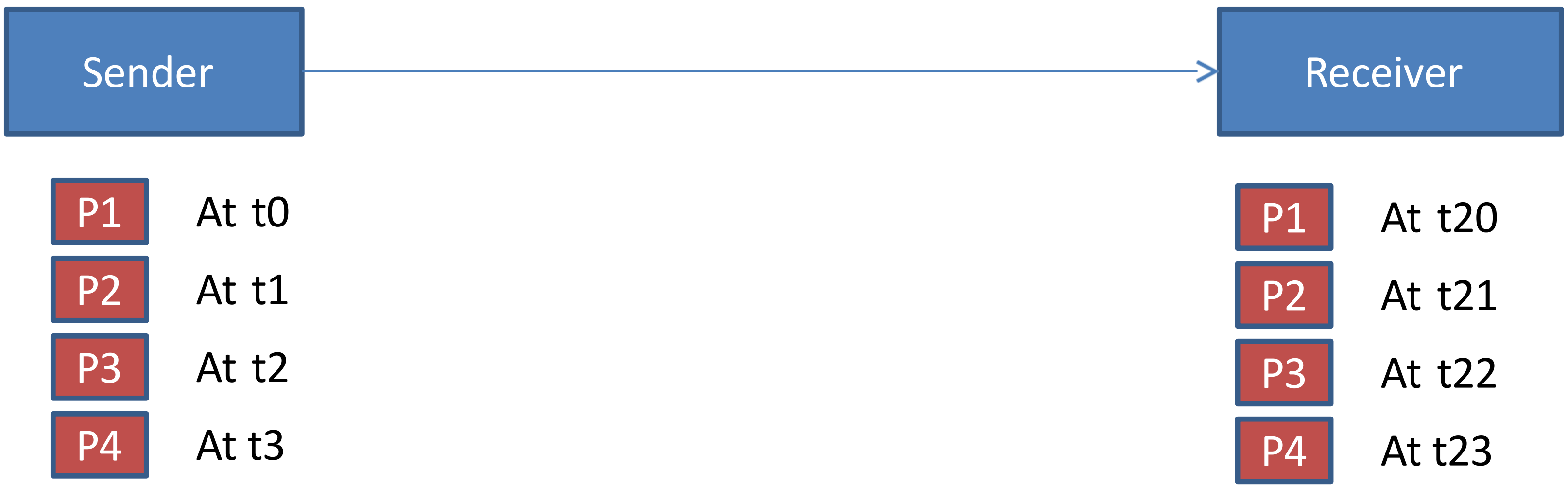
1. Constant transmission rate is important for audio/video
2. The variation in the packet arrival time is called "jitter"
3. Examples: (a) high jitter, 20 msec - 30 msec (Not Acceptable)
   (b) low jitter, 24.5 msec – 25.5 msec (Acceptable)



(a) High jitter. (b) Low jitter

5. Jitter Control

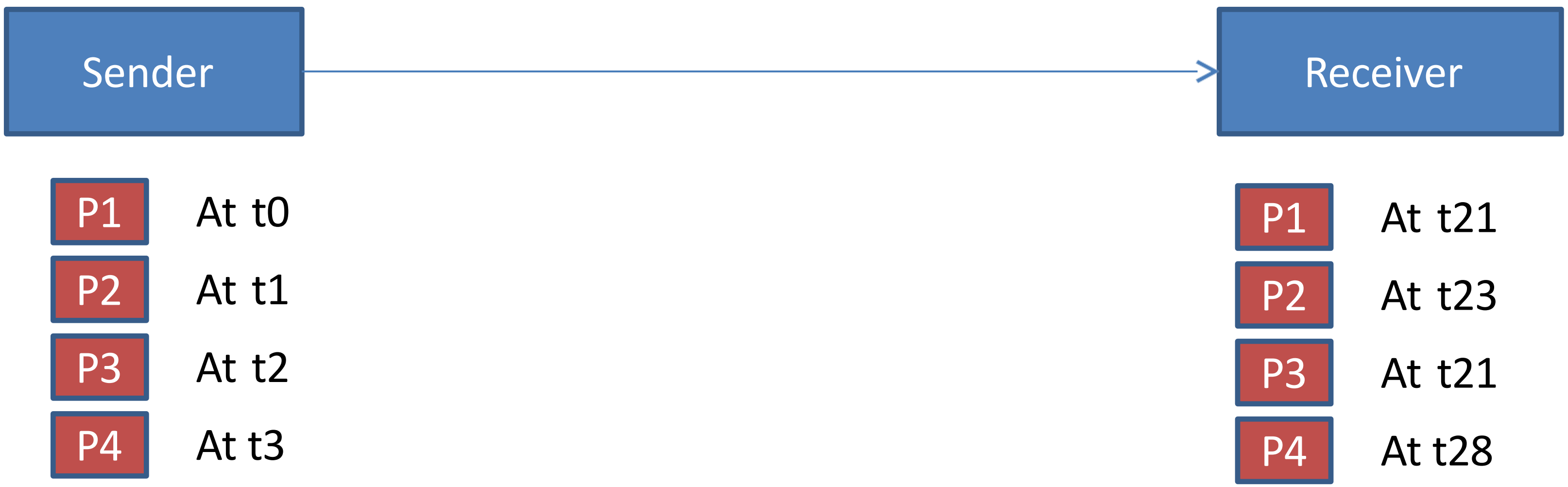
Different delays between packets cause jittering



all have the same delay, 20 units of time, so no jitter

5. Jitter Control

Different delays between packets cause jitter



have different delays: 21,22, 19, and 25, so there is jitter

5. Jitter Control

How to control Jitter?

When a packet arrives check to see how much the packet is behind or ahead of its schedule

- store this info. in the packet and update at each hop.
- **behind schedule ?** Transmit immediately
- **ahead of schedule ?** hold it just long enough to get it back on schedule. Then transmit.

5. Jitter Control

Note :

- In some applications, such as video on demand, jitter can be eliminated by buffering at the receiver.
- For real time applications (Internet telephony and videoconferencing) , buffering delay is not acceptable

Datagram Congestion Control Protocol (DCCP)

- DCCP is a transport layer protocol designed to provide congestion-controlled, unreliable message delivery for real-time applications.
- Limitations of existing protocols:
- TCP → provides congestion control but with reliability (ACKs, retransmissions) → not suitable for real-time (voice/video).
- UDP → provides fast, unreliable delivery but has no congestion control, causing network congestion and unfair bandwidth usage.
- DCCP fills the gap by providing:
- Congestion control (like TCP),Unreliable delivery (like UDP)
- Suitable for real-time multimedia (VoIP, online gaming, streaming)

DCCP Features

1. Congestion Control Support

DCCP includes built-in mechanisms for congestion control and supports several congestion control **profiles**, such as:

- **TCP-like congestion control**
- **TCP-Friendly Rate Control (TFRC)**

Applications can choose the profile based on their needs.

2. Unreliable, Message-Oriented Delivery

- No retransmissions
- No sequencing
- No delivery guarantees

This ensures **minimal delay**, essential for multimedia applications.

DCCP Features

3. Connection-Oriented

Even though it is unreliable, DCCP still establishes a connection (like TCP) to negotiate:

•Congestion control method

•Features

•Options

4. Packet Types

DCCP uses several control packets:

•**DCCP-Request** – connection initiation

•**DCCP-Response** – connection acceptance

DCCP-Data – data packets

•DCCP-Ack – acknowledgements for congestion control

•DCCP-Close / CloseReq – terminate connectio